

13a_ood_controlled_blend_test

April 28, 2026

1 13a — OOD-Controlled Blend Test

This notebook tests the **first extrapolation-control idea** only:

1. Detect OOD test scenarios using nearest-neighbor distance to the training set.
2. Compare three forward submission candidates:
 - `prediction_submission.csv` — conservative ExtraTrees final model.
 - `prediction_submission_hybrid_mlp_distance_w_0.65.csv` — aggressive MLP-distance hybrid.
 - a controlled blend between the two.
3. Create several blend candidates and inspect their test-set distributions.
4. Optionally create a new candidate submission file.

The goal is **not** to overwrite the official submission immediately. The goal is to decide whether a safer midpoint between ExtraTrees and the MLP-distance hybrid is better than choosing one extreme.

1.1 1. Imports and paths

```
[1]: from pathlib import Path
import json
import numpy as np
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
from sklearn.neighbors import NearestNeighbors

pd.set_option('display.max_columns', None)
pd.set_option('display.width', 180)

PROJECT_ROOT = Path.cwd().parent if Path.cwd().name == 'notebooks' else Path.
    cwd()
FORWARD_DIR = PROJECT_ROOT / 'data' / 'raw' / 'forward_prediction'
INVERSE_DIR = PROJECT_ROOT / 'data' / 'raw' / 'inverse_design'
OUTPUTS_DIR = PROJECT_ROOT / 'outputs'
SUBMISSIONS_DIR = OUTPUTS_DIR / 'submissions'
REPORTS_DIR = PROJECT_ROOT / 'reports'

for p in [SUBMISSIONS_DIR, REPORTS_DIR]:
    p.mkdir(parents=True, exist_ok=True)
```

```

print('Project root:', PROJECT_ROOT)
print('Forward dir:', FORWARD_DIR)
print('Submissions dir:', SUBMISSIONS_DIR)
print('Reports dir:', REPORTS_DIR)

```

Project root: /home/alouiyaz/projects/boom-challenge-ejecta-prediction
Forward dir: /home/alouiyaz/projects/boom-challenge-ejecta-prediction/data/raw/forward_prediction
Submissions dir: /home/alouiyaz/projects/boom-challenge-ejecta-prediction/outputs/submissions
Reports dir: /home/alouiyaz/projects/boom-challenge-ejecta-prediction/reports

1.2 2. Load train/test inputs and candidate submissions

```

[2]: input_cols = [
    'energy', 'angle_rad', 'coupling', 'strength',
    'porosity', 'gravity', 'atmosphere', 'shape_factor',
]

target_cols = [
    'P80', 'fines_frac', 'oversize_frac',
    'R95', 'R50_fines', 'R50_oversize',
]

fragmentation_cols = ['P80', 'fines_frac', 'oversize_frac']
distance_cols = ['R95', 'R50_fines', 'R50_oversize']

raw_train = pd.read_csv(FORWARD_DIR / 'train.csv')[input_cols]
raw_test = pd.read_csv(FORWARD_DIR / 'test.csv')[input_cols]
y = pd.read_csv(FORWARD_DIR / 'train_labels.csv')[target_cols]

base_path = SUBMISSIONS_DIR / 'prediction_submission.csv'
hybrid_path = SUBMISSIONS_DIR / 'prediction_submission_hybrid_mlp_distance_w_0.
↳65.csv'

if not base_path.exists():
    raise FileNotFoundError(f'Missing base submission: {base_path}')
if not hybrid_path.exists():
    raise FileNotFoundError(f'Missing hybrid submission: {hybrid_path}')

base = pd.read_csv(base_path)
hybrid = pd.read_csv(hybrid_path)

print('Raw train:', raw_train.shape)
print('Raw test:', raw_test.shape)
print('Base submission:', base.shape, base_path)

```

```
print('Hybrid submission:', hybrid.shape, hybrid_path)
```

```
display(raw_train.head())
```

```
display(raw_test.head())
```

```
display(base.head())
```

```
display(hybrid.head())
```

Raw train: (2930, 8)

Raw test: (492, 8)

Base submission: (492, 7) /home/alouiyaz/projects/boom-challenge-ejecta-prediction/outputs/submissions/prediction_submission.csv

Hybrid submission: (492, 7) /home/alouiyaz/projects/boom-challenge-ejecta-prediction/outputs/submissions/prediction_submission_hybrid_mlp_distance_w_0.65.csv

	energy	angle_rad	coupling	strength	porosity	gravity	atmosphere	
↪shape_factor								
0	3.826405	0.818303	0.861258	1.305809	0.337215	3.71	0.781263	0.
↪784028								
1	2.828754	1.193036	0.561245	3.494501	0.058029	1.62	0.136205	0.
↪922737								
2	3.068907	0.605872	0.948860	1.366386	0.315632	3.71	0.774704	0.
↪954922								
3	2.700574	1.073708	0.713705	3.599419	0.033062	1.62	0.144204	0.
↪932911								
4	3.484022	0.863568	1.237205	1.996742	0.278207	9.81	0.414620	1.
↪260855								

	energy	angle_rad	coupling	strength	porosity	gravity	atmosphere	
↪shape_factor								
0	4.387076	0.937830	1.305063	1.864599	0.162194	1.38	0.100630	0.
↪870897								
1	4.259426	0.848536	0.720038	1.803294	0.265631	4.91	0.708493	1.
↪188009								
2	4.384792	1.020636	1.003443	1.172195	0.114929	1.02	0.292809	0.
↪957242								
3	4.580126	0.888316	0.501471	2.721400	0.228540	1.02	0.787507	1.
↪365608								
4	3.607383	1.086271	1.170230	2.079364	0.184185	4.91	0.237429	0.
↪880140								

	scenario_id	P80	fines_frac	oversize_frac	R95	R50_fines	
↪R50_oversize							
0	0	120.852192	0.120046	0.208373	726.196054	704.966598	
↪312.487130							
1	1	113.668032	0.048219	0.158274	204.758075	220.793490	
↪102.212871							
2	2	149.740914	0.121343	0.384067	809.085123	808.873827	
↪348.096898							

3	3	162.227452	0.008732	0.543184	531.683980	592.186388	↵
↵	277.513349						
4	4	137.978607	0.047156	0.357337	197.063393	243.395743	↵
↵	100.912752						
	scenario_id	P80	fines_frac	oversize_frac	R95	R50_fines	↵
	↵R50_oversize						
0	0	120.852192	0.120046	0.208373	1064.238127	1012.805657	↵
↵	448.529033						
1	1	113.668032	0.048219	0.158274	180.340190	191.185816	↵
↵	88.486206						
2	2	149.740914	0.121343	0.384067	1306.856464	1316.118523	↵
↵	541.277606						
3	3	162.227452	0.008732	0.543184	731.377793	891.581710	↵
↵	418.480874						
4	4	137.978607	0.047156	0.357337	152.512030	142.451662	↵
↵	60.233150						

1.3 3. Submission format validation

Before blending, validate that both candidate files have the correct challenge format.

```
[3]: expected_cols = ['scenario_id'] + target_cols

def validate_submission(df, name):
    assert df.shape == (len(raw_test), 7), f'{name}: wrong shape {df.shape}'
    assert df.columns.tolist() == expected_cols, f'{name}: wrong columns'
    assert df['scenario_id'].tolist() == list(range(len(raw_test))), f'{name}: ↵
    ↵scenario_id must be 0-based index'
    assert np.isfinite(df.drop(columns=['scenario_id']).values).all(), f'{name}:
    ↵contains non-finite values'
    assert (df[['P80', 'R95', 'R50_fines', 'R50_oversize']] >= 0).all().all(), ↵
    ↵f'{name}: negative size/distance'
    assert ((df[['fines_frac', 'oversize_frac']] >= 0) & (df[['fines_frac', ↵
    ↵'oversize_frac']] <= 1)).all().all(), f'{name}: fraction outside [0, 1]'
    print(f'{name}: format OK')

validate_submission(base, 'base')
validate_submission(hybrid, 'hybrid')
```

```
base: format OK
hybrid: format OK
```

1.4 4. OOD detection with nearest-neighbor distance

We compute the distance of each test scenario to the training set in normalized input space.

A larger value means the test scenario is farther from the training distribution and therefore more exposed to extrapolation risk.

```
[4]: def compute_ood_distances(X_train_raw, X_query_raw, input_cols, n_neighbors=5):
    scaler = MinMaxScaler()
    train_scaled = scaler.fit_transform(X_train_raw[input_cols])
    query_scaled = scaler.transform(X_query_raw[input_cols])

    nn = NearestNeighbors(n_neighbors=n_neighbors)
    nn.fit(train_scaled)
    distances, indices = nn.kneighbors(query_scaled)

    return pd.DataFrame({
        'nearest_train_distance': distances[:, 0],
        'mean_5nn_distance': distances.mean(axis=1),
    })

test_ood = compute_ood_distances(raw_train, raw_test, input_cols, n_neighbors=5)
train_ood = compute_ood_distances(raw_train, raw_train, input_cols,
    ↪n_neighbors=6)

# For train-to-train, the nearest neighbor is itself with distance 0.
# Use mean_5nn_distance as a robust baseline; it includes self if using
    ↪query=train,
# but still provides a useful internal scale. We also compute a conservative
    ↪threshold from test distribution.
ood_threshold_train95 = np.percentile(train_ood['mean_5nn_distance'], 95)
ood_threshold_test75 = np.percentile(test_ood['mean_5nn_distance'], 75)

print('Train 95% OOD threshold:', round(ood_threshold_train95, 4))
print('Test 75% OOD threshold:', round(ood_threshold_test75, 4))
print('Test OOD using train95:', int((test_ood['mean_5nn_distance'] >
    ↪ood_threshold_train95).sum()), '/', len(test_ood))
print('Test far quartile using test75:', int((test_ood['mean_5nn_distance'] >
    ↪ood_threshold_test75).sum()), '/', len(test_ood))

test_ood = test_ood.reset_index().rename(columns={'index': 'scenario_id'})
test_ood['ood_bin'] = pd.qcut(test_ood['mean_5nn_distance'], q=4,
    ↪labels=['Q1_close', 'Q2', 'Q3', 'Q4_far'])
test_ood.to_csv(REPORTS_DIR / 'ood_distance_test_diagnostics.csv', index=False)

display(test_ood.describe().T)
display(test_ood.sort_values('mean_5nn_distance', ascending=False).head(20))
```

Train 95% OOD threshold: 0.2643

Test 75% OOD threshold: 0.7268

Test OOD using train95: 490 / 492

Test far quartile using test75: 123 / 492

	count	mean	std	min	25%	
↪ 50%	75%	max				

scenario_id	492.0	245.500000	142.172431	0.000000	122.750000	245.
↳500000	368.250000	491.000000				
nearest_train_distance	492.0	0.560674	0.149600	0.182295	0.435030	0.
↳580499	0.678606	0.889880				
mean_5nn_distance	492.0	0.617684	0.143318	0.234674	0.506642	0.
↳628117	0.726834	0.947235				

	scenario_id	nearest_train_distance	mean_5nn_distance	ood_bin
310	310	0.833301	0.947235	Q4_far
478	478	0.885638	0.918664	Q4_far
51	51	0.889880	0.911174	Q4_far
168	168	0.816546	0.903057	Q4_far
108	108	0.836186	0.902173	Q4_far
447	447	0.859306	0.896924	Q4_far
304	304	0.828850	0.895867	Q4_far
445	445	0.815027	0.891793	Q4_far
273	273	0.786785	0.891013	Q4_far
234	234	0.864652	0.888289	Q4_far
342	342	0.853755	0.882081	Q4_far
473	473	0.748515	0.878658	Q4_far
76	76	0.841188	0.870589	Q4_far
364	364	0.713797	0.869412	Q4_far
236	236	0.799806	0.869388	Q4_far
439	439	0.788228	0.868270	Q4_far
232	232	0.802410	0.860804	Q4_far
180	180	0.796829	0.860740	Q4_far
278	278	0.787044	0.856478	Q4_far
255	255	0.792770	0.854067	Q4_far

1.5 5. Compare base vs hybrid prediction shifts

The hybrid model improved CV on distance targets, but it extrapolated more strongly on the test set. Here we quantify how much it moves the distance predictions relative to the conservative ExtraTrees final model.

```
[5]: shift = test_ood.copy()
for col in target_cols:
    shift[f'{col}_base'] = base[col].values
    shift[f'{col}_hybrid'] = hybrid[col].values
    shift[f'{col}_diff'] = hybrid[col].values - base[col].values
    shift[f'{col}_abs_diff'] = np.abs(shift[f'{col}_diff'])

print('Correlation between OOD distance and absolute prediction shift:')
for col in distance_cols:
    corr = shift['mean_5nn_distance'].corr(shift[f'{col}_abs_diff'])
    print(f'{col}: {corr:.4f}')

summary_rows = []
```

```

for col in distance_cols:
    diff = shift[f'{col}_diff']
    summary_rows.append({
        'target': col,
        'base_mean': base[col].mean(),
        'hybrid_mean': hybrid[col].mean(),
        'mean_diff': diff.mean(),
        'median_abs_diff': diff.abs().median(),
        'p90_abs_diff': diff.abs().quantile(0.90),
        'p95_abs_diff': diff.abs().quantile(0.95),
        'max_abs_diff': diff.abs().max(),
        'n_abs_diff_gt_100': int((diff.abs() > 100).sum()),
        'n_abs_diff_gt_250': int((diff.abs() > 250).sum()),
    })

shift_summary = pd.DataFrame(summary_rows)
display(shift_summary)

cols = ['scenario_id', 'mean_5nn_distance', 'ood_bin']
for col in distance_cols:
    cols += [f'{col}_base', f'{col}_hybrid', f'{col}_diff', f'{col}_abs_diff']

display(shift.sort_values('R95_abs_diff', ascending=False)[cols].head(25))

```

Correlation between OOD distance and absolute prediction shift:

R95: 0.1651

R50_fines: 0.1613

R50_oversize: 0.1119

	target	base_mean	hybrid_mean	mean_diff	median_abs_diff	
↪	p90_abs_diff	p95_abs_diff	max_abs_diff	n_abs_diff_gt_100	n_abs_diff_gt_250	
0	R95	298.530443	367.205990	68.675547	35.872930	332.
↪	709830	451.172075	810.228242		116	63
1	R50_fines	330.556280	393.496650	62.940369	57.340424	345.
↪	972356	507.756459	871.151783		132	96
2	R50_oversize	146.295647	178.261386	31.965738	23.330167	160.
↪	194100	191.370530	347.170558		104	6
	scenario_id	mean_5nn_distance	ood_bin	R95_base	R95_hybrid	R95_diff
↪	R95_abs_diff	R50_fines_base	R50_fines_hybrid	R50_fines_diff		
↪	R50_fines_abs_diff	\				
479	479	0.849991	Q4_far	772.512211	1582.740453	810.228242
↪	810.228242	726.864392	1598.016175	871.151783		871.
↪	151783					
250	250	0.661713	Q3	928.510613	1654.971472	726.460860
↪	726.460860	832.051264	1560.287127	728.235864		728.
↪	235864					

404	404	0.825827	Q4_far	659.167320	1346.470250	687.302930
↪	687.302930	691.440059		1393.488106	702.048047	702.
↪	048047					
185	185	0.820441	Q4_far	621.450940	1268.344065	646.893125
↪	646.893125	674.079218		1329.993948	655.914730	655.
↪	914730					
318	318	0.728975	Q4_far	843.278073	1458.464597	615.186524
↪	615.186524	752.642424		1213.298448	460.656024	460.
↪	656024					
45	45	0.690118	Q3	799.704741	1389.471151	589.766409
↪	589.766409	728.858959		1384.757121	655.898162	655.
↪	898162					
349	349	0.715488	Q3	863.535296	1448.974395	585.439099
↪	585.439099	793.167985		1351.168717	558.000733	558.
↪	000733					
416	416	0.805646	Q4_far	791.038525	1363.597193	572.558667
↪	572.558667	745.131087		1326.259158	581.128071	581.
↪	128071					
206	206	0.805969	Q4_far	665.586738	1235.478708	569.891971
↪	569.891971	670.724899		1313.953165	643.228266	643.
↪	228266					
433	433	0.796502	Q4_far	636.843751	1200.339968	563.496217
↪	563.496217	689.823890		1300.121477	610.297586	610.
↪	297586					
409	409	0.765744	Q4_far	746.924660	1298.804837	551.880176
↪	551.880176	738.628346		1335.068040	596.439694	596.
↪	439694					
137	137	0.661154	Q3	577.699067	1112.808167	535.109100
↪	535.109100	652.489029		1242.940828	590.451799	590.
↪	451799					
180	180	0.860740	Q4_far	644.996029	1170.509784	525.513756
↪	525.513756	661.158479		1204.041032	542.882554	542.
↪	882554					
370	370	0.742770	Q4_far	626.678171	1148.710641	522.032470
↪	522.032470	681.891123		1263.965710	582.074588	582.
↪	074588					
342	342	0.882081	Q4_far	768.171926	1286.764520	518.592594
↪	518.592594	745.992498		1307.018979	561.026482	561.
↪	026482					
216	216	0.684800	Q3	804.598322	1303.070499	498.472177
↪	498.472177	772.012573		1261.871647	489.859074	489.
↪	859074					
417	417	0.733363	Q4_far	605.523705	1103.827729	498.304025
↪	498.304025	666.746964		1221.474217	554.727252	554.
↪	727252					

2	2	0.703846	Q3	809.085123	1306.856464	497.771341
↪	497.771341	808.873827		1316.118523	507.244696	507.
	↪244696					
445	445	0.891793	Q4_far	645.873011	1142.724282	496.851271
↪	496.851271	682.394905		1083.221646	400.826742	400.
	↪826742					
71	71	0.615945	Q2	961.033661	1449.475728	488.442067
↪	488.442067	844.358079		1368.416873	524.058794	524.
	↪058794					
148	148	0.569900	Q2	734.776059	1209.019435	474.243376
↪	474.243376	742.195296		1284.322424	542.127128	542.
	↪127128					
320	320	0.591562	Q2	974.018339	1440.994622	466.976283
↪	466.976283	836.175703		1366.474507	530.298803	530.
	↪298803					
66	66	0.723692	Q3	694.804836	1160.082275	465.277438
↪	465.277438	723.638590		1232.020537	508.381947	508.
	↪381947					
362	362	0.633605	Q3	818.736057	1276.611531	457.875475
↪	457.875475	789.895718		1304.703899	514.808180	514.
	↪808180					
41	41	0.612163	Q2	821.981341	1274.118671	452.137330
↪	452.137330	766.885338		1288.197154	521.311816	521.
	↪311816					

	R50_oversize_base	R50_oversize_hybrid	R50_oversize_diff	
	↪R50_oversize_abs_diff			
479	330.522469	677.693027	347.170558	347.
	↪170558			
250	401.941819	665.885509	263.943690	263.
	↪943690			
404	293.656241	579.982499	286.326257	286.
	↪326257			
185	276.527204	545.439803	268.912599	268.
	↪912599			
318	356.904899	543.319538	186.414639	186.
	↪414639			
45	335.080871	591.396736	256.315865	256.
	↪315865			
349	352.698989	556.360146	203.661157	203.
	↪661157			
416	333.704629	543.986339	210.281710	210.
	↪281710			
206	291.218519	563.238543	272.020024	272.
	↪020024			

433	285.368497	530.356159	244.987663	244.
↳987663				
409	330.305516	569.122132	238.816616	238.
↳816616				
137	267.487372	505.896005	238.408633	238.
↳408633				
180	278.701253	500.080146	221.378892	221.
↳378892				
370	285.162367	518.587199	233.424832	233.
↳424832				
342	326.881557	550.783488	223.901930	223.
↳901930				
216	338.397999	516.564999	178.167000	178.
↳167000				
417	274.461041	489.125578	214.664537	214.
↳664537				
2	348.096898	541.277606	193.180709	193.
↳180709				
445	290.112589	468.151335	178.038746	178.
↳038746				
71	400.526096	580.179201	179.653105	179.
↳653105				
148	331.311958	552.088674	220.776716	220.
↳776716				
320	405.072718	600.763064	195.690346	195.
↳690346				
66	300.005576	502.883297	202.877722	202.
↳877722				
362	359.534313	571.232194	211.697881	211.
↳697881				
41	346.872281	552.905521	206.033239	206.
↳033239				

1.6 6. Shift by OOD quartile

This section checks whether the hybrid model mainly changes predictions for the farthest OOD cases. If yes, the hybrid is reacting to OOD. If no, it may be unstable even inside familiar regions.

```
[6]: quartile_summary = []
for col in distance_cols:
    grouped = shift.groupby('ood_bin').agg(
        n=('scenario_id', 'size'),
        mean_ood=('mean_5nn_distance', 'mean'),
        base_mean=(f'{col}_base', 'mean'),
        hybrid_mean=(f'{col}_hybrid', 'mean'),
        mean_diff=(f'{col}_diff', 'mean'),
        mean_abs_diff=(f'{col}_abs_diff', 'mean'),
```

```

        max_abs_diff=(f'{col}_abs_diff', 'max'),
    ).reset_index()
    grouped['target'] = col
    quartile_summary.append(grouped)

quartile_summary = pd.concat(quartile_summary, ignore_index=True)
display(quartile_summary[['target', 'ood_bin', 'n', 'mean_ood', 'base_mean', '
    ↪ 'hybrid_mean', 'mean_diff', 'mean_abs_diff', 'max_abs_diff']])

```

	target	ood_bin	n	mean_ood	base_mean	hybrid_mean	mean_diff	
	↪ mean_abs_diff	max_abs_diff						
0	R95	Q1_close	123	0.428039	313.032453	361.108014	48.075562	↪
	↪ 69.925521	433.969553						
1	R95	Q2	123	0.567561	308.171464	365.670681	57.499217	↪
	↪ 89.716358	488.442067						
2	R95	Q3	123	0.681224	299.220215	369.759895	70.539679	↪
	↪ 110.021791	726.460860						
3	R95	Q4_far	123	0.793915	273.697639	372.285368	98.587729	↪
	↪ 129.937007	810.228242						
4	R50_fines	Q1_close	123	0.428039	348.194732	412.305827	64.111095	↪
	↪ 93.608933	530.601511						
5	R50_fines	Q2	123	0.567561	341.312161	402.555356	61.243195	↪
	↪ 115.414085	542.127128						
6	R50_fines	Q3	123	0.681224	325.716762	383.304048	57.587286	↪
	↪ 136.090314	728.235864						
7	R50_fines	Q4_far	123	0.793915	307.001466	375.821367	68.819902	↪
	↪ 154.797221	871.151783						
8	R50_oversize	Q1_close	123	0.428039	159.608852	195.081247	35.472395	↪
	↪ 45.615009	196.368379						
9	R50_oversize	Q2	123	0.567561	152.614310	184.562572	31.948263	↪
	↪ 51.345904	220.776716						
10	R50_oversize	Q3	123	0.681224	142.026614	168.392380	26.365766	↪
	↪ 55.533616	263.943690						
11	R50_oversize	Q4_far	123	0.793915	130.932813	165.009343	34.076530	↪
	↪ 63.561218	347.170558						

1.7 7. Create controlled blend candidates

We create several candidates between the conservative base and the aggressive hybrid.

Only distance targets are blended. Fragmentation targets remain exactly from the final ExtraTrees model.

Formula:

$$\text{controlled_distance} = (1 - \alpha) * \text{base_distance} + \alpha * \text{hybrid_distance}$$

Where:

- $\alpha = 0.00 \rightarrow$ pure ExtraTrees final.

- $\alpha = 1.00 \rightarrow$ pure hybrid MLP-distance model.
- $\alpha = 0.50 \rightarrow$ midpoint, equivalent to around 67.5% ExtraTrees + 32.5% MLP internally.

```
[7]: def create_controlled_blend(base_df, hybrid_df, alpha, distance_cols,
    ↪ fragmentation_cols):
    blend = base_df.copy()
    for col in fragmentation_cols:
        blend[col] = base_df[col].values
    for col in distance_cols:
        blend[col] = (1 - alpha) * base_df[col].values + alpha * hybrid_df[col].
    ↪ values
        blend[col] = np.clip(blend[col], 0, None)
    return blend

alphas = [0.25, 0.40, 0.50, 0.60, 0.75]
blend_files = {}
blend_summaries = []

for alpha in alphas:
    candidate = create_controlled_blend(base, hybrid, alpha, distance_cols,
    ↪ fragmentation_cols)
    validate_submission(candidate, f'controlled_blend_alpha_{alpha:.2f}')

    out_path = SUBMISSIONS_DIR /
    ↪ f'prediction_submission_controlled_blend_alpha_{alpha:.2f}.csv'
    candidate.to_csv(out_path, index=False)
    blend_files[alpha] = out_path

    row = {'alpha': alpha, 'path': str(out_path)}
    for col in distance_cols:
        row[f'{col}_mean'] = candidate[col].mean()
        row[f'{col}_median'] = candidate[col].median()
        row[f'{col}_p75'] = candidate[col].quantile(0.75)
        row[f'{col}_p95'] = candidate[col].quantile(0.95)
        row[f'{col}_max'] = candidate[col].max()
    blend_summaries.append(row)

blend_summary_df = pd.DataFrame(blend_summaries)
display(blend_summary_df)
print('Saved blend candidates:')
for alpha, path in blend_files.items():
    print(alpha, path)
```

```
controlled_blend_alpha_0.25: format OK
controlled_blend_alpha_0.40: format OK
controlled_blend_alpha_0.50: format OK
controlled_blend_alpha_0.60: format OK
controlled_blend_alpha_0.75: format OK
```

	alpha	path	R95_mean	
	R95_median	R95_p75	R95_p95	R95_max R50_fines_mean
	R50_fines_median	R50_fines_p75	\	
0	0.25	/home/alouiyaz/projects/boom-challenge-ejecta-...	315.699329	168.
	481906	525.714033	902.956827	1110.125827 346.291373 200.025927
		604.380533		
1	0.40	/home/alouiyaz/projects/boom-challenge-ejecta-...	326.000661	163.
	317604	534.597301	971.246236	1219.094956 355.732428 193.092528
		619.928522		
2	0.50	/home/alouiyaz/projects/boom-challenge-ejecta-...	332.868216	159.
	304984	543.370131	1020.374924	1291.741042 362.026465 187.567113
		629.519979		
3	0.60	/home/alouiyaz/projects/boom-challenge-ejecta-...	339.735771	156.
	283265	553.933893	1058.307792	1364.387128 368.320502 181.989295
		648.782200		
4	0.75	/home/alouiyaz/projects/boom-challenge-ejecta-...	350.037103	151.
	958390	567.002821	1114.887662	1473.356257 377.761557 172.665834
		661.556107		

	R50_fines_p95	R50_fines_max	R50_oversize_mean	R50_oversize_median	
	R50_oversize_p75	R50_oversize_p95	R50_oversize_max		
0	867.336532	1014.110230	154.287082	87.607904	
	281.195952	382.894259	467.927742		
1	935.068908	1123.345609	159.081943	84.674456	
	291.157898	410.354963	507.519295		
2	987.186304	1196.169196	162.278517	82.701016	
	297.701791	428.443965	533.913664		
3	1040.551927	1268.992782	165.475090	80.895347	
	303.416589	447.827905	560.308033		
4	1117.427642	1380.228229	170.269951	77.755055	
	316.797186	474.755744	599.899587		

Saved blend candidates:

0.25 /home/alouiyaz/projects/boom-challenge-ejecta-prediction/outputs/submissions/prediction_submission_controlled_blend_alpha_0.25.csv

0.4 /home/alouiyaz/projects/boom-challenge-ejecta-prediction/outputs/submissions/prediction_submission_controlled_blend_alpha_0.40.csv

0.5 /home/alouiyaz/projects/boom-challenge-ejecta-prediction/outputs/submissions/prediction_submission_controlled_blend_alpha_0.50.csv

0.6 /home/alouiyaz/projects/boom-challenge-ejecta-prediction/outputs/submissions/prediction_submission_controlled_blend_alpha_0.60.csv

0.75 /home/alouiyaz/projects/boom-challenge-ejecta-prediction/outputs/submissions/prediction_submission_controlled_blend_alpha_0.75.csv

1.8 8. Distribution comparison: base vs hybrid vs controlled blends

```
[8]: models = {'base_extratrees': base, 'hybrid_w_0.65': hybrid}
for alpha, path in blend_files.items():
    models[f'controlled_alpha_{alpha:.2f}'] = pd.read_csv(path)

rows = []
for name, df in models.items():
    for col in distance_cols:
        rows.append({
            'model': name,
            'target': col,
            'mean': df[col].mean(),
            'std': df[col].std(),
            'min': df[col].min(),
            'median': df[col].median(),
            'p75': df[col].quantile(0.75),
            'p90': df[col].quantile(0.90),
            'p95': df[col].quantile(0.95),
            'max': df[col].max(),
        })

distribution_comparison = pd.DataFrame(rows)
display(distribution_comparison)

distribution_comparison.to_csv(REPORTS_DIR /
    'controlled_blend_distribution_comparison.csv', index=False)
```

	model	target	mean	std	min	
median	p75	p90	p95	max		
0	base_extratrees	R95	298.530443	241.158254	48.776910	177.
453510	508.553148	648.373860	795.225069	996.810828		
1	base_extratrees	R50_fines	330.556280	235.993585	64.210862	213.
098175	580.026235	670.231723	738.834486	873.066616		
2	base_extratrees	R50_oversize	146.295647	108.429543	28.072745	91.
697836	263.430771	291.156824	332.525738	414.330461		
3	hybrid_w_0.65	R95	367.205990	387.867101	30.070443	144.
903024	589.193077	1038.646374	1235.550442	1654.971472		
4	hybrid_w_0.65	R50_fines	393.496650	403.381768	38.428605	156.
017613	681.769264	1002.674852	1240.745131	1598.016175		
5	hybrid_w_0.65	R50_oversize	178.261386	178.035458	13.158020	71.
683292	328.076953	448.575919	517.474989	677.693027		
6	controlled_alpha_0.25	R95	315.699329	276.157022	46.649578	168.
481906	525.714033	733.610742	902.956827	1110.125827		
7	controlled_alpha_0.25	R50_fines	346.291373	275.564557	62.038436	200.
025927	604.380533	753.443940	867.336532	1014.110230		

8	controlled_alpha_0.25	R50_oversize	154.287082	125.016964	26.713766	87.
	607904	281.195952	332.956577	382.894259	467.927742	
9	controlled_alpha_0.40	R95	326.000661	297.850464	44.676256	163.
	317604	534.597301	803.535054	971.246236	1219.094956	
10	controlled_alpha_0.40	R50_fines	355.732428	300.276096	60.734980	193.
	092528	619.928522	795.946348	935.068908	1123.345609	
11	controlled_alpha_0.40	R50_oversize	159.081943	135.309464	25.257191	84.
	674456	291.157898	359.831137	410.354963	507.519295	
12	controlled_alpha_0.50	R95	332.868216	312.535487	42.727333	159.
	304984	543.370131	853.733507	1020.374924	1291.741042	
13	controlled_alpha_0.50	R50_fines	362.026465	317.050084	59.866010	187.
	567113	629.519979	832.473411	987.186304	1196.169196	
14	controlled_alpha_0.50	R50_oversize	162.278517	142.278867	23.346041	82.
	701016	297.701791	370.204587	428.443965	533.913664	
15	controlled_alpha_0.60	R95	339.735771	327.370761	40.464124	156.
	283265	553.933893	893.808428	1058.307792	1364.387128	
16	controlled_alpha_0.60	R50_fines	368.320502	334.021720	57.984025	181.
	989295	648.782200	867.848157	1040.551927	1268.992782	
17	controlled_alpha_0.60	R50_oversize	165.475090	149.320403	21.308437	80.
	895347	303.416589	385.745983	447.827905	560.308033	
18	controlled_alpha_0.75	R95	350.037103	349.863535	36.566494	151.
	958390	567.002821	937.400350	1114.887662	1473.356257	
19	controlled_alpha_0.75	R50_fines	377.761557	359.788499	51.949421	172.
	665834	661.556107	921.373850	1117.427642	1380.228229	
20	controlled_alpha_0.75	R50_oversize	170.269951	159.997095	18.252031	77.
	755055	316.797186	409.462683	474.755744	599.899587	

1.9 9. OOD-adaptive blend candidate

This version uses a **smaller hybrid weight for close test points** and a **larger hybrid weight for far OOD points**.

This is experimental. It tries to use MLP extrapolation only where extrapolation is most likely needed.

```
[9]: def create_ood_adaptive_blend(base_df, hybrid_df, ood_df, distance_cols,
    fragmentation_cols, min_alpha=0.20, max_alpha=0.60):
    blend = base_df.copy()
    for col in fragmentation_cols:
        blend[col] = base_df[col].values

    scores = ood_df['mean_5nn_distance'].values
    lo = np.percentile(scores, 25)
    hi = np.percentile(scores, 90)
    scaled = (scores - lo) / (hi - lo + 1e-12)
    scaled = np.clip(scaled, 0, 1)
    alpha = min_alpha + (max_alpha - min_alpha) * scaled
```

```

    for col in distance_cols:
        blend[col] = (1 - alpha) * base_df[col].values + alpha * hybrid_df[col].
↪values
        blend[col] = np.clip(blend[col], 0, None)

    blend['adaptive_alpha_debug'] = alpha
    return blend

adaptive = create_ood_adaptive_blend(
    base, hybrid, test_ood,
    distance_cols=distance_cols,
    fragmentation_cols=fragmentation_cols,
    min_alpha=0.20,
    max_alpha=0.60,
)

adaptive_alpha = adaptive[['scenario_id', 'adaptive_alpha_debug']].copy()
adaptive_submission = adaptive.drop(columns=['adaptive_alpha_debug'])
validate_submission(adaptive_submission, 'ood_adaptive_blend')

adaptive_path = SUBMISSIONS_DIR /
↪'prediction_submission_ood_adaptive_blend_alpha_0.20_0.60.csv'
adaptive_submission.to_csv(adaptive_path, index=False)
adaptive_alpha.to_csv(REPORTS_DIR / 'ood_adaptive_blend_alpha_debug.csv',
↪index=False)

print('Saved:', adaptive_path)
display(adaptive_alpha.describe().T)
display(adaptive_submission.head())
display(adaptive_submission[distance_cols].describe().T)

```

ood_adaptive_blend: format OK

Saved: /home/alouiyaz/projects/boom-challenge-ejecta-prediction/outputs/submissions/prediction_submission_ood_adaptive_blend_alpha_0.20_0.60.csv

		count	mean	std	min	25%	50%
↪	75% max						
scenario_id		492.0	245.500000	142.172431	0.0	122.750000	245.500000
↪	368.250000 491.0						
adaptive_alpha_debug		492.0	0.374075	0.147049	0.2	0.200375	0.368838
↪	0.506043 0.6						
scenario_id	P80	fines_frac	oversize_frac		R95	R50_fines	
↪	R50_oversize						
0	0	120.852192	0.120046	0.208373	879.903826	844.941056	
↪	374.345400						

1	1	113.668032	0.048219	0.158274	199.874498	214.871955	↵
↵	99.467538						
2	2	149.740914	0.121343	0.384067	1045.074616	1049.354563	↵
↵	439.682358						
3	3	162.227452	0.008732	0.543184	571.622743	652.065452	↵
↵	305.706854						
4	4	137.978607	0.047156	0.357337	179.767194	204.206170	↵
↵	85.119689						
	count	mean	std	min	25%	50%	↵
↵	75%	max					
R95	492.0	327.133550	302.132929	40.464124	97.864575	164.663765	↵
↵	529.450783	1258.649156					
R50_fines	492.0	354.400257	303.531438	58.184816	123.299538	195.370609	↵
↵	605.401438	1249.555462					
R50_oversize	492.0	158.137348	135.776475	21.308437	51.190738	85.665521	↵
↵	282.864169	538.824804					

1.10 10. Final candidate comparison table

Use this table to decide which candidate has a reasonable balance between stability and OOD extrapolation.

```
[10]: final_models = dict(models)
final_models['ood_adaptive_alpha_0.20_0.60'] = adaptive_submission

rows = []
for name, df in final_models.items():
    row = {'model': name}
    for col in distance_cols:
        row[f'{col}_mean'] = df[col].mean()
        row[f'{col}_median'] = df[col].median()
        row[f'{col}_p75'] = df[col].quantile(0.75)
        row[f'{col}_p95'] = df[col].quantile(0.95)
        row[f'{col}_max'] = df[col].max()
    rows.append(row)

candidate_comparison = pd.DataFrame(rows)
display(candidate_comparison)
candidate_comparison.to_csv(REPORTS_DIR /
    ↵ 'final_forward_candidate_distribution_comparison.csv', index=False)

print('Candidate files created:')
print('Base official:', base_path)
print('Hybrid aggressive:', hybrid_path)
for alpha, path in blend_files.items():
    print(f'Controlled alpha {alpha:.2f}:', path)
print('OOD adaptive:', adaptive_path)
```

	model	R95_mean	R95_median	R95_p75	R95_p95	
↪	R95_max	R50_fines_mean	R50_fines_median	R50_fines_p75	R50_fines_p95	↪
↪	R50_fines_max	\				
0	base_extratrees	298.530443	177.453510	508.553148	795.225069	↪
↪	996.810828	330.556280	213.098175	580.026235	738.834486	↪
↪	873.066616					
1	hybrid_w_0.65	367.205990	144.903024	589.193077	1235.550442	↪
↪	1654.971472	393.496650	156.017613	681.769264	1240.745131	↪
↪	1598.016175					
2	controlled_alpha_0.25	315.699329	168.481906	525.714033	902.956827	↪
↪	1110.125827	346.291373	200.025927	604.380533	867.336532	↪
↪	1014.110230					
3	controlled_alpha_0.40	326.000661	163.317604	534.597301	971.246236	↪
↪	1219.094956	355.732428	193.092528	619.928522	935.068908	↪
↪	1123.345609					
4	controlled_alpha_0.50	332.868216	159.304984	543.370131	1020.374924	↪
↪	1291.741042	362.026465	187.567113	629.519979	987.186304	↪
↪	1196.169196					
5	controlled_alpha_0.60	339.735771	156.283265	553.933893	1058.307792	↪
↪	1364.387128	368.320502	181.989295	648.782200	1040.551927	↪
↪	1268.992782					
6	controlled_alpha_0.75	350.037103	151.958390	567.002821	1114.887662	↪
↪	1473.356257	377.761557	172.665834	661.556107	1117.427642	↪
↪	1380.228229					
7	ood_adaptive_alpha_0.20_0.60	327.133550	164.663765	529.450783	976.641967	↪
↪	1258.649156	354.400257	195.370609	605.401438	932.897330	↪
↪	1249.555462					
	R50_oversize_mean	R50_oversize_median	R50_oversize_p75	R50_oversize_p95		↪
↪	R50_oversize_max					
0	146.295647	91.697836	263.430771	332.525738		↪
↪	414.330461					
1	178.261386	71.683292	328.076953	517.474989		↪
↪	677.693027					
2	154.287082	87.607904	281.195952	382.894259		↪
↪	467.927742					
3	159.081943	84.674456	291.157898	410.354963		↪
↪	507.519295					
4	162.278517	82.701016	297.701791	428.443965		↪
↪	533.913664					
5	165.475090	80.895347	303.416589	447.827905		↪
↪	560.308033					
6	170.269951	77.755055	316.797186	474.755744		↪
↪	599.899587					
7	158.137348	85.665521	282.864169	417.213472		↪
↪	538.824804					

Candidate files created:

Base official: /home/alouiyaz/projects/boom-challenge-ejecta-prediction/outputs/submissions/prediction_submission.csv

Hybrid aggressive: /home/alouiyaz/projects/boom-challenge-ejecta-prediction/outputs/submissions/prediction_submission_hybrid_mlp_distance_w_0.65.csv

Controlled alpha 0.25: /home/alouiyaz/projects/boom-challenge-ejecta-prediction/outputs/submissions/prediction_submission_controlled_blend_alpha_0.25.csv

Controlled alpha 0.40: /home/alouiyaz/projects/boom-challenge-ejecta-prediction/outputs/submissions/prediction_submission_controlled_blend_alpha_0.40.csv

Controlled alpha 0.50: /home/alouiyaz/projects/boom-challenge-ejecta-prediction/outputs/submissions/prediction_submission_controlled_blend_alpha_0.50.csv

Controlled alpha 0.60: /home/alouiyaz/projects/boom-challenge-ejecta-prediction/outputs/submissions/prediction_submission_controlled_blend_alpha_0.60.csv

Controlled alpha 0.75: /home/alouiyaz/projects/boom-challenge-ejecta-prediction/outputs/submissions/prediction_submission_controlled_blend_alpha_0.75.csv

OOD adaptive: /home/alouiyaz/projects/boom-challenge-ejecta-prediction/outputs/submissions/prediction_submission_ood_adaptive_blend_alpha_0.20_0.60.csv

1.11 11. Decision notes

Recommended interpretation:

- If the hybrid model creates very large distance values for many test scenarios, it is aggressive and risky.
- If the base ExtraTrees model appears too conservative for far-OOD test points, a controlled blend can be a safer compromise.
- The most useful candidate is usually the one that increases distance predictions moderately without creating extreme maximum values.

Do **not** overwrite outputs/submissions/prediction_submission.csv inside this notebook.

After reviewing the distribution comparison, if we decide to submit one candidate, copy it manually to the official filename.

Example:

```
cp outputs/submissions/prediction_submission_controlled_blend_alpha_0.50.csv outputs/submissions/
```

or

```
cp outputs/submissions/prediction_submission_ood_adaptive_blend_alpha_0.20_0.60.csv outputs/submissions/
```